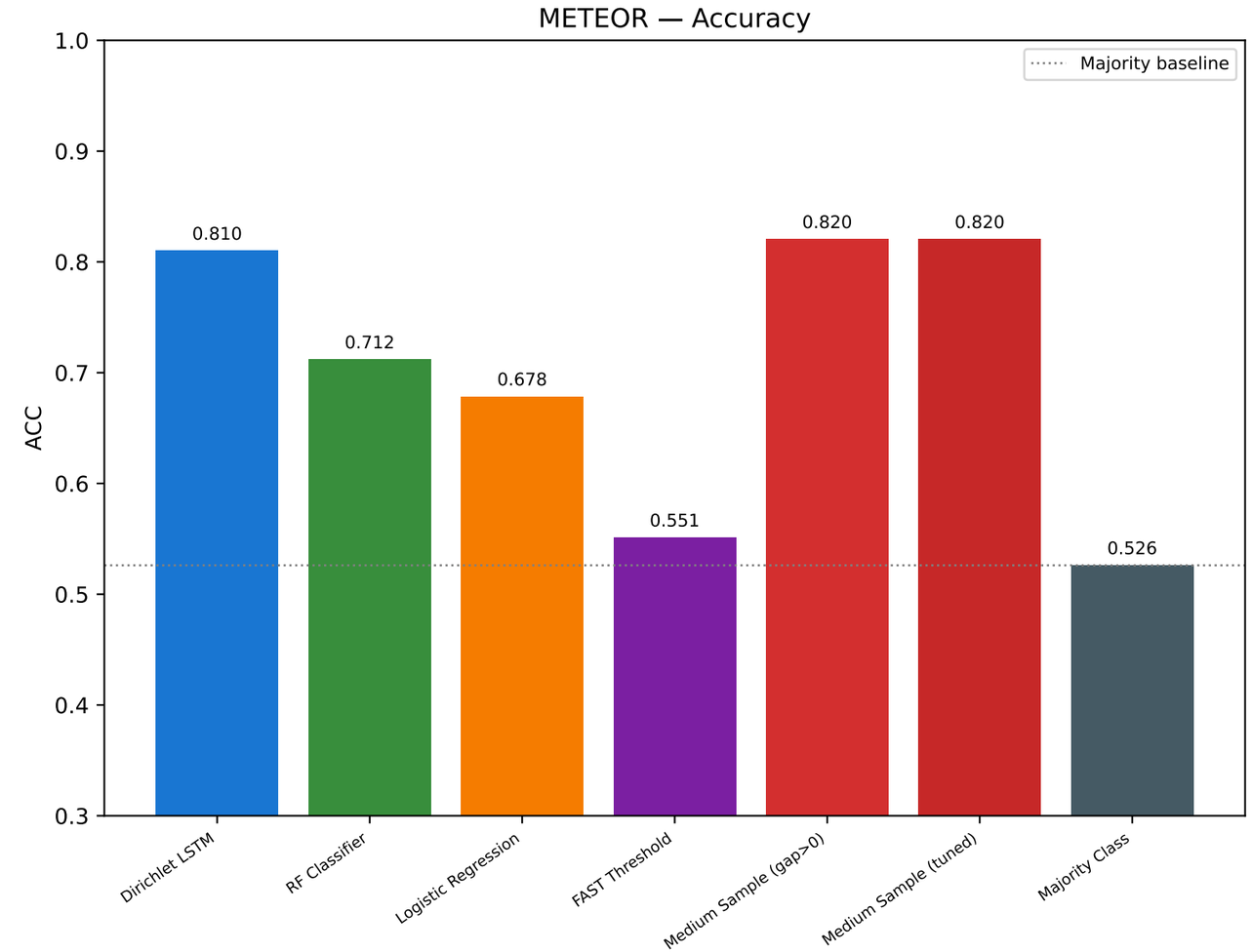
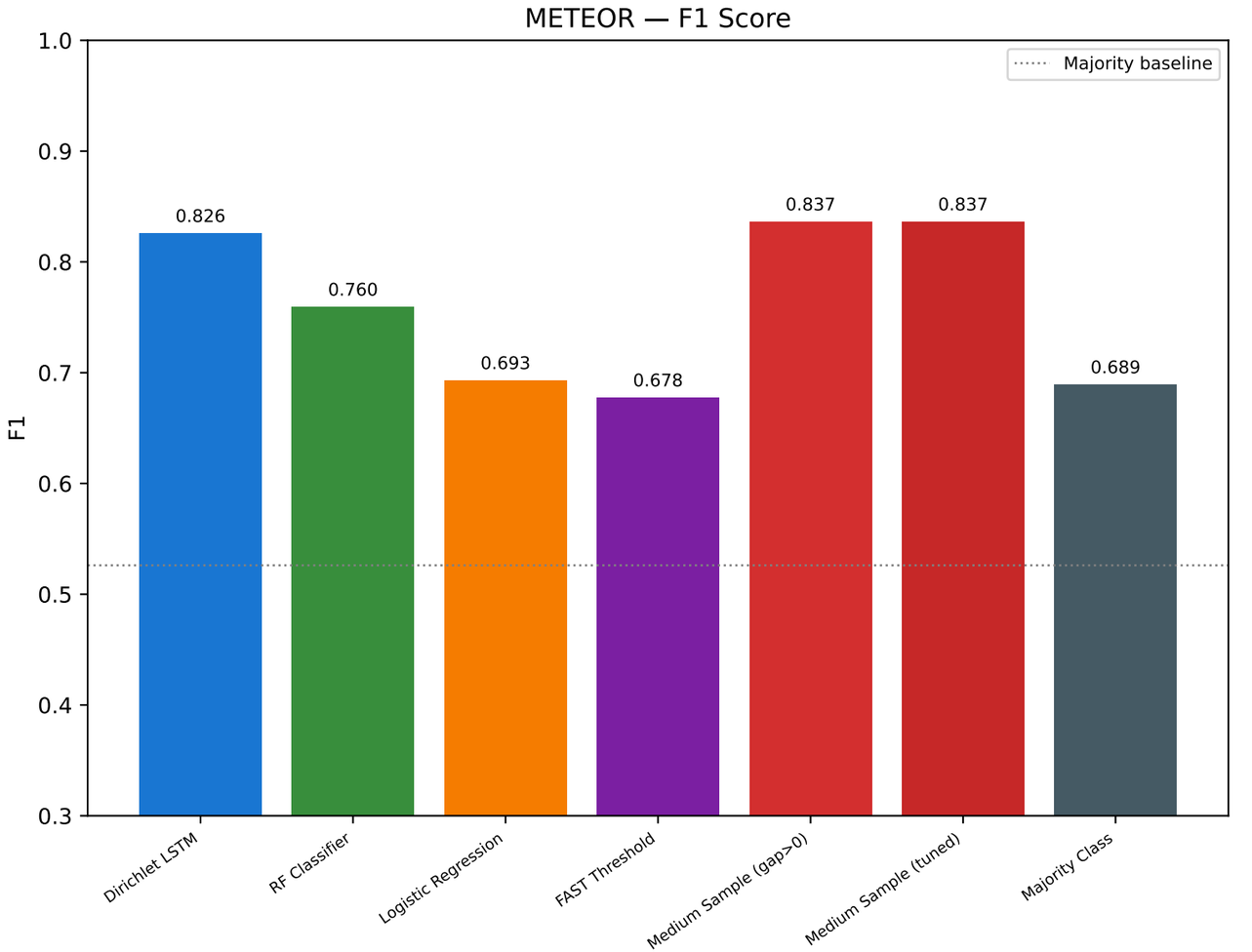
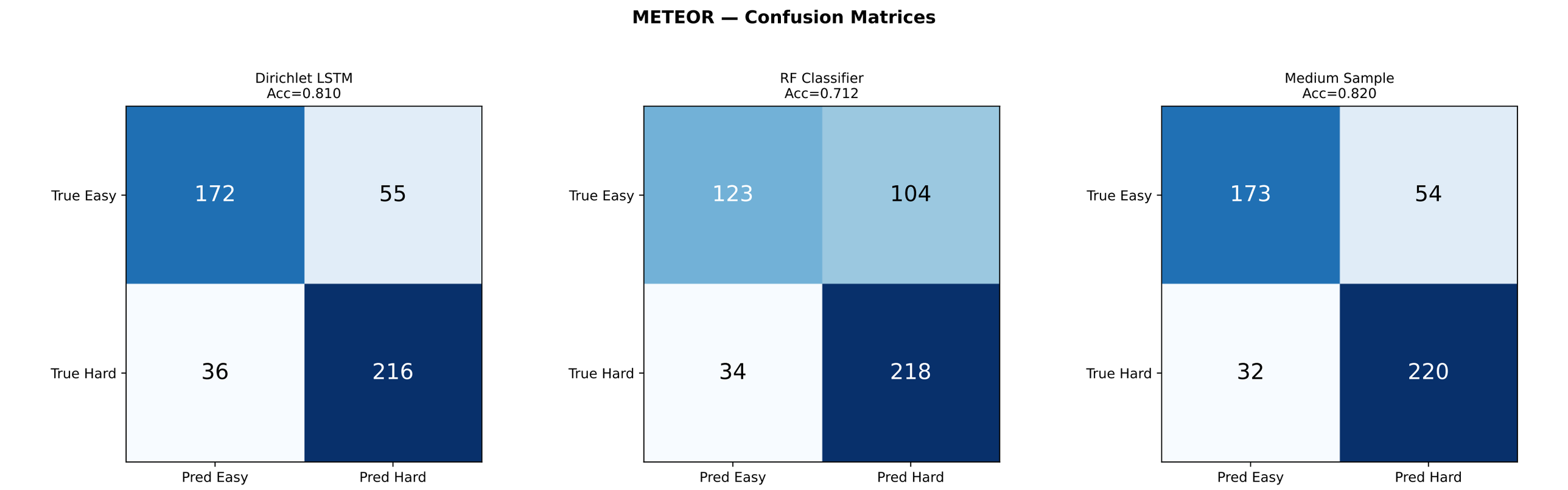
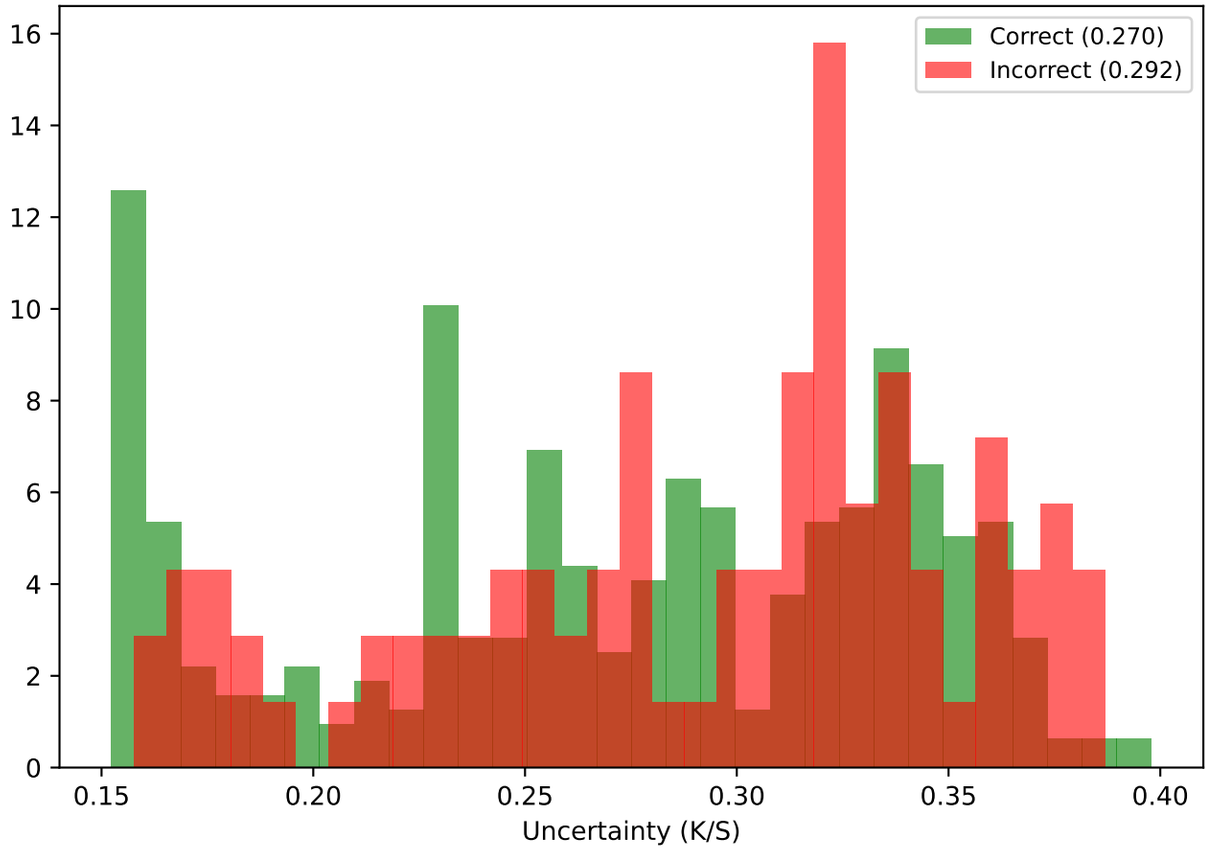
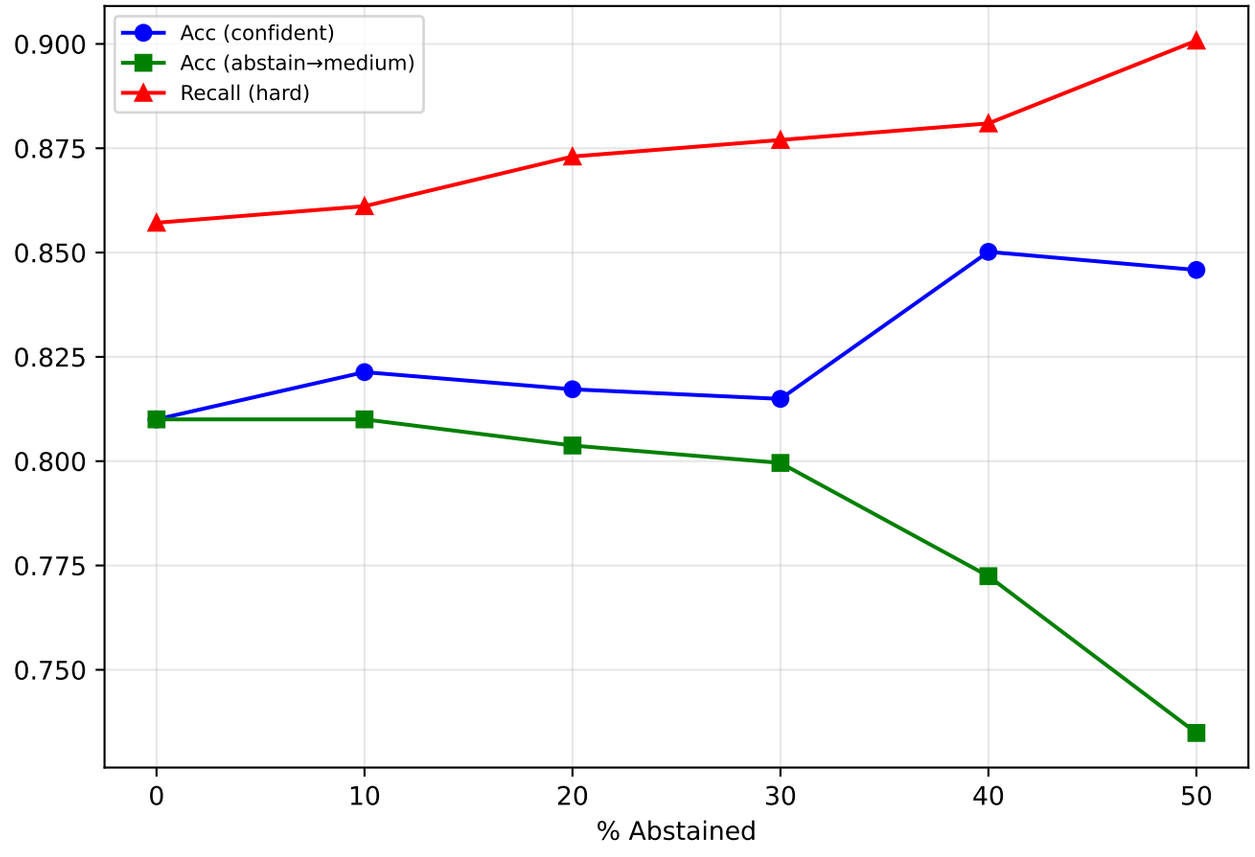
METEOR Dataset — Dirichlet LSTM vs All Baselines

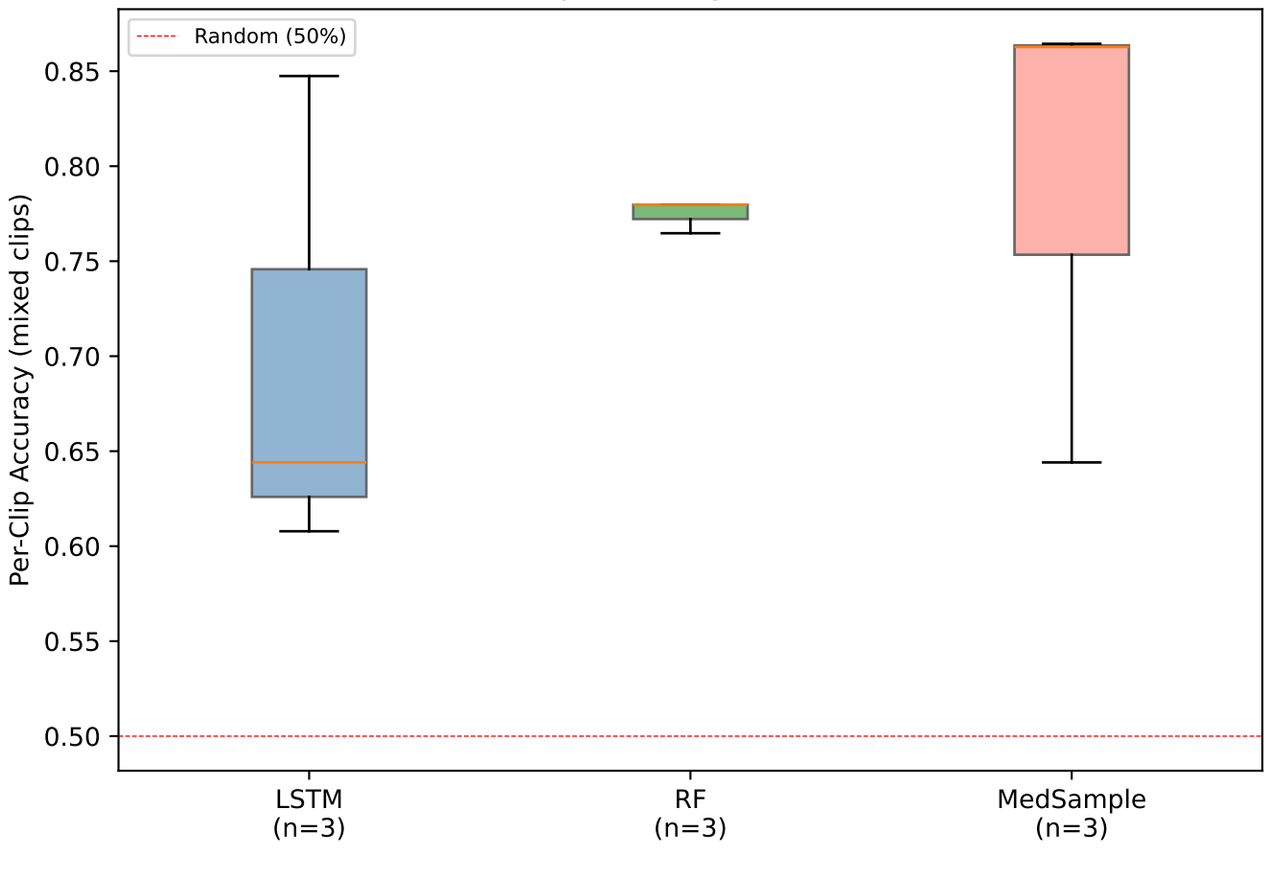


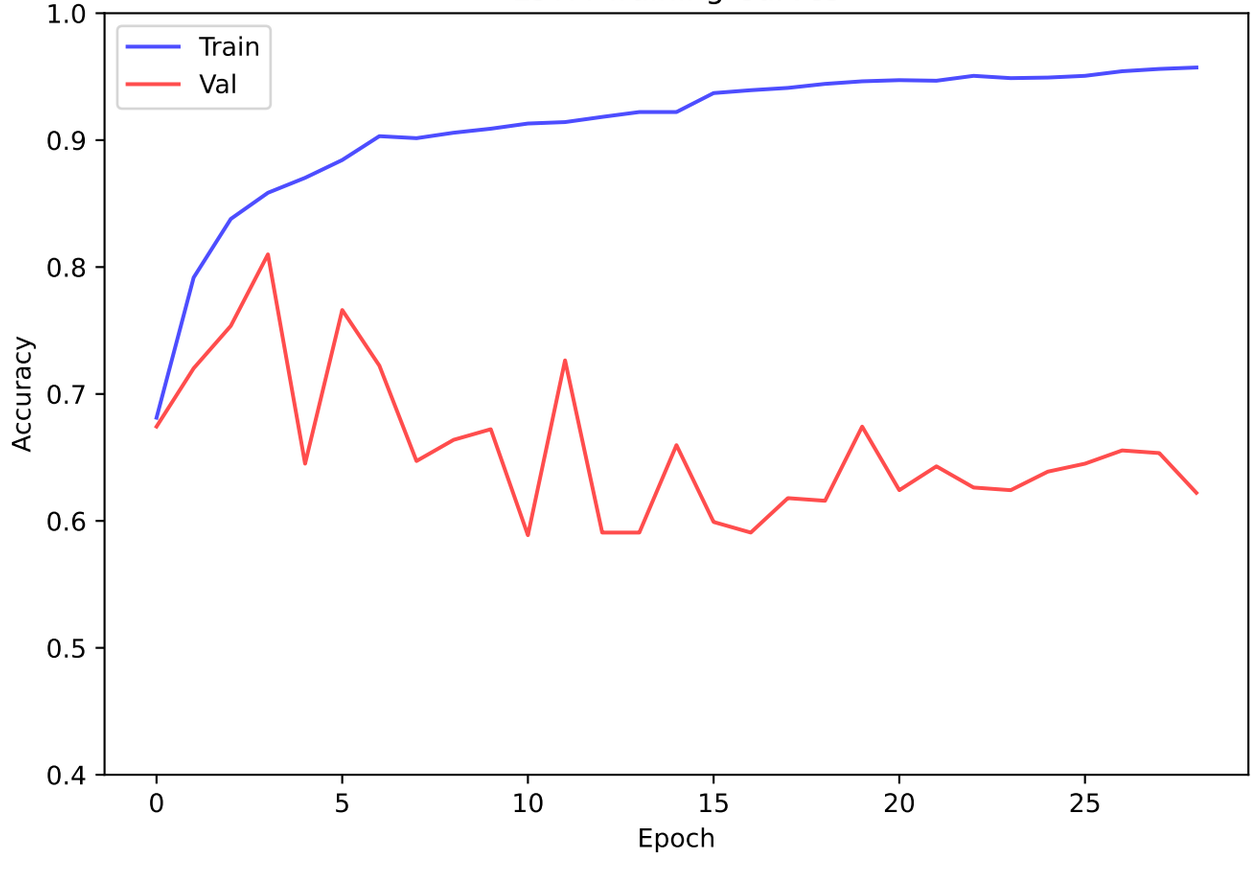


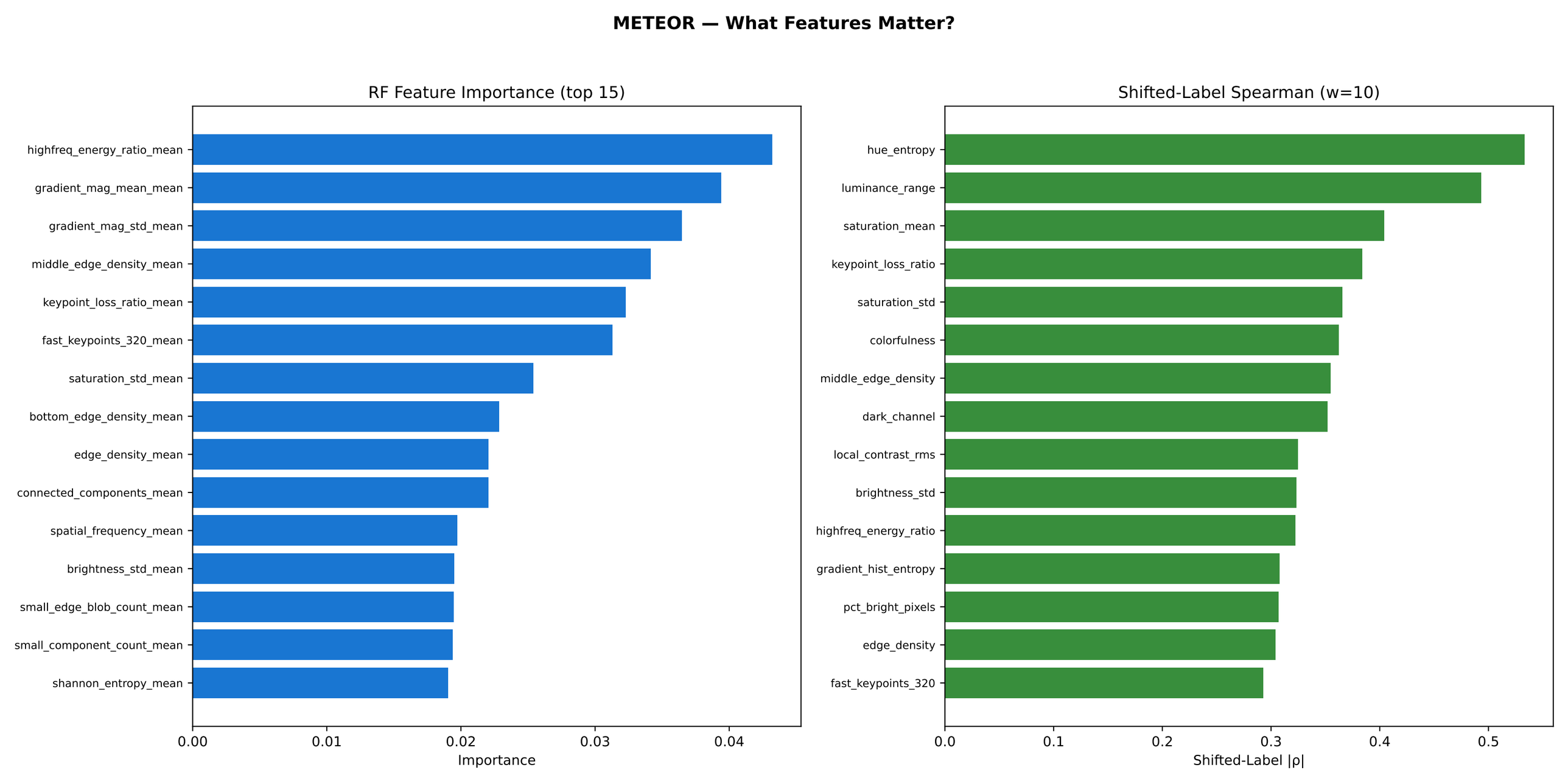

METEOR — Confusion Matrices
Dirichlet LSTM
Acc=0.810
True Easy
True Hard
172
55
36
216
Pred Easy
Pred Hard
RF Classifier
Acc=0.712
True Easy
True Hard
123
104
34
218
Pred Easy
Pred Hard
Medium Sample
Acc=0.820
True Easy
True Hard
173
54
32
220
Pred Easy
Pred Hard


Uncertainty: Correct vs Incorrect
Correct (0.270)
Incorrect (0.292)
Uncertainty (K/S)


Uncertainty-Based Abstention
Acc (confident)
Acc (abstain→medium)
Recall (hard)
% Abstained


Per-Clip Accuracy Distribution
Random (50%)
Per-Clip Accuracy (mixed clips)
LSTM (n=3)
RF (n=3)
MedSample (n=3)


LSTM Training Curves
Train
Val
Accuracy
Epoch


METEOR — What Features Matter?
RF Feature Importance (top 15)
highfreq_energy_ratio_mean
gradient_mag_mean_mean
gradient_mag_std_mean
middle_edge_density_mean
keypoint_loss_ratio_mean
fast_keypoints_320_mean
saturation_std_mean
bottom_edge_density_mean
edge_density_mean
connected_components_mean
spatial_frequency_mean
brightness_std_mean
small_edge_blob_count_mean
small_component_count_mean
shannon_entropy_mean
0.00
0.01
0.02
0.03
0.04
Importance
Shifted-Label Spearman (w=10)
hue_entropy
luminance_range
saturation_mean
keypoint_loss_ratio
saturation_std
colorfulness
middle_edge_density
dark_channel
local_contrast_rms
brightness_std
highfreq_energy_ratio
gradient_hist_entropy
pct_bright_pixels
edge_density
fast_keypoints_320
0.0
0.1
0.2
0.3
0.4
0.5
Shifted-Label |ρ|

```
================================================================================
METEOR DATASET — FULL EVALUATION SUMMARY
================================================================================

Dataset:     49 clips, ~10 fps, 27,574 frames, 41 features
Task:        predict mean(fn_coco_nano[t+10:t+20]) >= 3.1
             from 10 frames (1 sec) of spatial features
Class split: 52.6% hard / 47.4% easy
Val set:     9 clips, 479 non-overlapping windows

Model                              Acc     F1   Prec    Rec    AUC
------------------------------------------------------------
Dirichlet LSTM                   0.810  0.826  0.797  0.857 0.861
RF Classifier                    0.712  0.760  0.677  0.865 0.804
Logistic Regression              0.678  0.693  0.696  0.690 0.715
FAST Threshold                   0.551  0.678  0.545  0.897   —
Medium Sample (gap>0)            0.820  0.837  0.803  0.873   —
Medium Sample (tuned)            0.820  0.837  0.803  0.873   —
Majority Class                   0.526  0.689  0.526  1.000   —

KEY COMPARISON:
  Dirichlet LSTM:     81.0% accuracy
  Medium-sampling:    82.0% accuracy
  Majority baseline:  52.6% accuracy

VERDICT:
  Medium-sampling beats the LSTM.
================================================================================
```